

CBSE Class 12 Chemistry (56/1/2) Question Paper 2026

General Instruction

Read the following instructions carefully and follow them :

- (i) This question paper contains 33 questions. All questions are compulsory.**
- (ii) This question paper is divided into five sections – Section A, B, C, D and E.**
- (iii) Section A – questions number 1 to 16 are multiple choice type questions. Each question carries 1 mark.**
- (iv) Section B – questions number 17 to 21 are very short answer type questions. Each question carries 2 marks.**
- (v) Section C – questions number 22 to 28 are short answer type questions. Each question carries 3 marks.**
- (vi) Section D – questions number 29 and 30 are case-based questions. Each question carries 4 marks.**
- (vii) Section E – questions number 31 to 33 are long answer type questions. Each question carries 5 marks.**
- (viii) There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the sections except Section – A.**

(ix) Kindly note that there is a separate question paper for Visually Impaired candidates.

(x) Use of calculator is NOT allowed.

If you want, I can also clean this into a proper table or bullet format for your article.

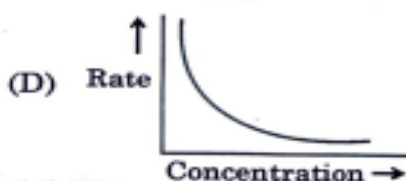
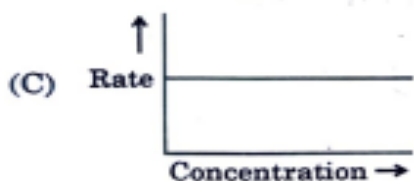
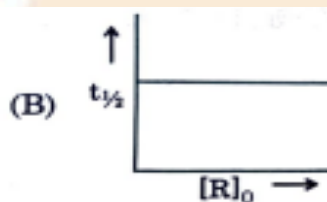
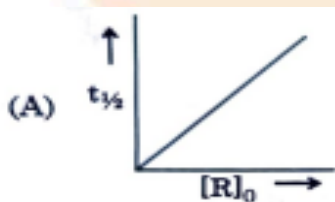
SECTION A

Questions No. 1 to 16 are Multiple Choice type questions carrying 1 mark each.

1. Which of the following is most basic?

- (A) Mn_2O_7
- (B) MnO_2
- (C) MnO
- (D) Mn_2O_3

2. Which of the following curve represents the first order reaction?



3. Which of the following solutions will have the highest osmotic pressure?

- (A) 0.1 M KCl
- (B) 0.1 M CaCl_2
- (C) 0.1 M Glucose
- (D) 0.1 M Urea

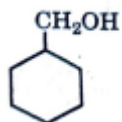
4. For a reaction : $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, the rate of reaction with respect to NH_3 is

- (A) $+ \frac{1}{3} \Delta[\text{NH}_3] / \Delta t$
- (B) $- \frac{1}{2} \Delta[\text{NH}_3] / \Delta t$
- (C) $+ \frac{1}{4} \Delta[\text{NH}_3] / \Delta t$
- (D) $+ \frac{1}{2} \Delta[\text{NH}_3] / \Delta t$

5. How many Faradays are required to reduce 1 mol of $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} in an acidic medium?

- (A) 2
- (B) 3
- (C) 6
- (D) 4

6. Which of the following Grignard Reagent will be used to prepare



when treated with methanal



7. The secondary valency of Pt in $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$ is

- (A) 6
- (B) 5
- (C) 4
- (D) 2

8. $(\text{CH}_3)_3\text{C} - \text{OC}_2\text{H}_5$ on reaction with HI gives

- (A) $(\text{CH}_3)_3\text{C} - \text{I}$ and $\text{C}_2\text{H}_5 - \text{I}$
- (B) $(\text{CH}_3)_3\text{C} - \text{OH}$ and $\text{C}_2\text{H}_5 - \text{I}$
- (C) $(\text{CH}_3)_3\text{C} - \text{I}$ and $\text{C}_2\text{H}_5 - \text{OH}$
- (D) $(\text{CH}_3)_3\text{C} - \text{OH}$ and $\text{C}_2\text{H}_5 - \text{OH}$

9. Aniline on direct nitration yields

- (A) 51%-ortho, 47%-para, 2%-meta derivatives
- (B) 51%-meta, 47%-ortho, 2%-para derivatives
- (C) 51%-para, 47%-meta, 2%-ortho derivatives
- (D) 51%-ortho, 47%-meta, 2%-para derivatives

10. Which of the following is 'not' true about enantiomers?

- (A) They have the same chemical reactivity.
- (B) They have the same specific rotation.

- (C) They have the same melting or boiling point.
- (D) They have the same refractive index.

11. An example of non-reducing sugar is

- (A) Glucose
- (B) Sucrose
- (C) Lactose
- (D) Maltose

12. Benzene diazonium chloride on reaction with phenol in weakly basic medium gives

- (A) Azobenzene
- (B) Benzene
- (C) Chlorobenzene
- (D) p-hydroxyazobenzene

For questions number 13 to 16, two statements are given – one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below :

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false, but Reason (R) is true.

13. Assertion (A): The presence of $-OH$ group in phenols directs the incoming group to ortho and para positions.

Reason (R): $-OH$ group in phenols activates the aromatic ring towards electrophilic substitution reaction.

14. Assertion (A): Actinoids show irregularities in their electronic configurations.

Reason (R): In actinoids 5f, 6d and 7s orbitals are of comparable energies.

15. Assertion (A): Components of azeotropes are easily separated by fractional distillation.

Reason (R): Components of an azeotrope have same composition in liquid and vapour phase.

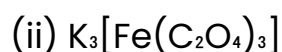
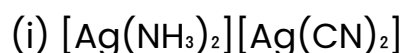
16. Assertion (A): The two strands of DNA are complementary to each other.

Reason (R): The hydrogen bonds are formed between specific pairs of bases.

SECTION – B

17. What type of deviation from Raoult's law is shown by mixture of ethanol and acetone? Give reason. What will happen to the boiling point of the solution on mixing ethanol and acetone?

18. (a) Write IUPAC names of the following coordination compounds :



OR

(b)

(i) Give a chemical test to show that $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$ and $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ are ionisation isomers.

(ii) What is meant by the 'Chelate effect'? Give an example.

19. Why are haloarenes less reactive towards nucleophilic substitution reaction? Give two reasons.

20. Following reaction takes place in one step :



How will the rate of above reaction change if the volume of the reaction vessel is decreased to one third of its original volume? Will there be any change in the order of reaction with the reduced volume?

21. Differentiate between the following :

- (i) Acidic amino acids and basic amino acids
- (ii) Nucleotide and Nucleoside

SECTION – C

22. Compound 'X' with molecular formula $\text{C}_4\text{H}_9\text{Br}$ reacts with aqueous KOH to give an alcohol. The rate of this reaction depends only on concentration of compound 'X'. When an optically active isomer 'Y' of the compound 'X' was treated with aqueous KOH solution, the rate of reaction was found to be dependent on concentration of compound 'Y' and aqueous KOH both.

- (a) Write down the structural formula of both 'X' and 'Y'.
 (b) Out of 'X' and 'Y', which one will undergo racemisation and why?
 (c) Out of 'X' and 'Y', which one will form product with inversion of configuration and why?

23. Write the reaction involved in the following :

- (a) Reimer-Tiemann reaction
 (b) Kolbe's reaction
 (c) Friedel-Crafts acylation of anisole

24. Give reasons for the following :

- (a) Carboxylic acids have higher boiling point than alcohols of comparable molecular masses.
 (b) Alpha (α) hydrogens of aldehydes and ketones are acidic in nature.
 (c) Nucleophilic addition of ammonia and its derivatives does not occur with carbonyl group in strongly acidic medium.

25. Define the following terms :

- (a) Anomers
 (b) Invert sugar
 (c) Glycosidic linkage

26. For the first order thermal decomposition reaction, following data was obtained :



S. No.	Time(s)	Total Pressure (atm)
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1	0	0.30
2	30	0.50

Calculate rate constant. **[Given : $\log 3 = 0.48$]**

27. (a) Answer the following :

- (i) Why is the Equilibrium Constant (K_c) related to E°_{cell} and not to E_{cell} ?
- (ii) Two metals 'A' and 'B' have standard electrode potential values of **-0.24 V** and **+0.80 V** respectively. Which of these will liberate hydrogen gas from dil. H_2SO_4 ?
- (iii) Write the cell reaction which occurs in lead storage battery when it is in charging.

OR

(b) What type of battery is Mercury cell? Why is it more advantageous than dry cell? Write overall reaction taking place in Mercury cell.

****28.** Calculate the boiling point of a solution containing 0.61 g of benzoic acid (Molar mass = 122 g mol^{-1}) in 5 g of CS_2 if benzoic acid dimerises to the extent of 88%. The boiling point and K_b of CS_2 are 46.2°C and $2.3 \text{ K kg mol}^{-1}$ respectively.

SECTION – D

The following questions are case-based questions. Each question has an internal choice and carries 4 (2 + 1 + 1) marks each. Read the passage carefully and answer the questions based on that follows :

29. The **Valence Bond Theory (VBT)** explains the formation, magnetic behaviour and geometry of coordination compounds. The **Crystal Field Theory (CFT)** of coordination compounds is based on the effect of different crystal fields (provided by the ligands taken as point charges), on the degeneracy of d-orbital energies of the central metal atom/ion. The splitting of the d-orbitals provides different electronic arrangements in strong and weak crystal fields.

Answer the following questions :

(a) In octahedral crystal field, energies of which d-orbitals will be raised when ligands approach the central metal atom/ion? Give reason in support of your answer.

(b) Using crystal field theory, write the electronic configuration of central metal atom/ion of the following :

(i) $[\text{CoF}_6]^{3-}$

(ii) $[\text{Co}(\text{NH}_3)_6]^{3+}$

[At. No. : Co = 27]

(c) $[\text{NiCl}_4]^{2-}$ is paramagnetic while $[\text{Ni}(\text{CO})_4]$ is diamagnetic though both are tetrahedral. Why?

[Atomic No. : Ni = 28]

OR

(c) Write hybridization and magnetic behaviour of the complex $[\text{Fe}(\text{CN})_6]^{3-}$.
[Atomic No. : Fe = 26]

30. The reaction of amines with mineral acids to form ammonium salts shows that these are basic in nature. Aliphatic amines are stronger bases than ammonia, whereas aromatic amines are weaker bases than ammonia. Aliphatic and aromatic primary and secondary amines react with acid chlorides, anhydrides and esters by nucleophilic substitution reaction. The main problem encountered during electrophilic substitution reactions to aromatic amines is that of their high reactivity. Substitution tends to occur at ortho- and para-positions. Hinsberg reagent is used for the identification and distinction between primary, secondary and tertiary amines. Aryldiazonium salts, usually obtained from arylamines, undergo replacement of the diazonium group with a variety of nucleophiles to provide advantageous methods for producing aryl halides, cyanides, phenols and arenes.

Answer the following questions :

30.

(a)

(i) Why $\text{CH}_3 - \text{NH}_2$ is a stronger base than $(\text{CH}_3)_3\text{N}$ in aqueous solution?

(ii) Write structural formulae of the compound **A** and **B** :



(b) A compound 'X' with molecular formula C_3H_9N reacts with Hinsberg reagent to give a product insoluble in alkali. Identify 'X'.

OR

(b) How can you convert aniline into benzonitrile?

(c) Why is $-NH_2$ group of aniline acetylated before carrying out nitration?

SECTION – E

31. (a) Calculate emf and ΔG for the following cell at 298 K :



Given :

$$E^\circ_{Mg^{2+}/Mg} = -2.37 V$$

$$E^\circ_{Ag^+/Ag} = +0.80 V$$

$$[1 F = 96500 C mol^{-1}, \log 10 = 1]$$

OR

(b) For the reaction :



Calculate emf of the cell at $25^\circ C$.

Given : $\Delta G^\circ = -43500 J mol^{-1}$

$$[\log 10 = 1, 1 F = 96500 C mol^{-1}]$$

32.

(a)

(i) An organic compound **(X)** has the molecular formula **C₅H₁₀O**. Draw structures for **(X)** if it :

(I) does not give Tollens' test but gives a positive iodoform test.

(II) does not give Tollen's test and iodoform test but undergoes Aldol condensation.

(III) undergoes Cannizzaro's reaction.

(ii) Show how each of the following compounds can be converted to benzoic acid :

(I) Acetophenone

(II) Ethyl benzene

OR

(b) Answer the following questions :

(i) Draw structure of the **2,4-dinitrophenyl hydrazone derivative of benzaldehyde**.

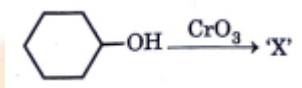
(ii) Arrange the following in increasing order of their reactivity towards HCN :

Di-tert. butyl ketone, Acetaldehyde, Acetone

(iii) Give a simple chemical test to distinguish between **benzoic acid** and **ethyl benzoate**.

(iv) Write the name of the reagent to convert **Ethanenitrile** to **Ethanal**.

(v) Draw the structure of 'X' in the following reaction :



33. (a) From the given data of E° values, answer the following questions :

E° (M^{2+}/M)	V	Cr	Mn	Fe	Co	Ni	Cu
	-1.1 8	-0.9 1	-1.1 8	-0.4 4	-0.2 8	-0.2 5	+0.3 4

(i) Why $E^\circ \text{Mn}^{2+}/\text{Mn}$ show irregular trend in the above values?

(ii) Why is $E^\circ \text{Cu}^{2+}/\text{Cu}$ value exceptionally positive?

(iii) Why $E^\circ \text{Mn}^{2+}/\text{Mn}$ value is highly negative?

(ii) Write the ionic equations for the oxidising action of potassium permanganate for its reaction with I^- in both acidic and alkaline solutions.

OR

(b) Answer the following questions :

(i) Name a member of the lanthanoid series

(I) which exhibits +4 oxidation state

(II) which exhibits +2 oxidation state.

(ii) Why transition metals act as good catalyst?

(iii) Why Cr has higher melting point than Mn?

(iv) What happens when acidic solution of potassium permanganate is allowed to stand for sometime? Give the equation involved. What is this type of reaction called?